KEYS

U4
TCA9554
I2C_SCL
I2C_SDA
KEY_INT_N
+3V3
GND
BTN_BACK
BTN_MENU
BTN_FN
BTN_MARK
BTN_PAGE
HAPTIC_EN
EXP_SPARE
EXP_SPARE
BACK_RAW
MENU_RAW
FN_RAW
MARK_RAW
D1
SP0504BAHT
C6
100n X8R
R8
10k
R9
10k
R10
100k
R12
10k
R13
10k
R14
10k
R15
10k
R16
10k
R17
100k
R18
100k
R19
100k
R20
100k
R21
100k
C8
10n C0G
C9
10n C0G
C10
10n C0G
C11
10n C0G
C12
10n C0G
SW1
BACK
SW2
MENU
SW3
FN
SW4
MARK
SW5
PAGE

AMP

U7
PAM8302A
AMP_INP
AMP_INN
AMP_SD
AMP_OUTP
AMP_OUTN
AMP_5V
GND
C22
100n X8R
R40
10k
R34
100k
C25
100n C0G
C26
100n C0G
R35
10k
R36
10k
C23
100p C0G
C24
100p C0G
C27
470p C0G 1%
C28
470p C0G 1%
FB1
220R@100M 1.4A
FB2
220R@100M 1.4A
J4
Speaker 4R 2.5W

FPC

J1
FPC 26p 0.5mm
C18
10u X7R 16V
C29
1u X8R 25V

SOS

U5
74AUP1G17
SOS_TMR
SOS_LONG
D3
SP0504BAHT
R28
10k
R29
1k
C16
10n C0G
R30
910k
C17
4.7u X8R 16V
C7
100n X8R
SW7
SOS
D5
1N4148WS

ENCODER

SW6
EC12D 15p/30det +push
ENC_A_RAW
ENC_B_RAW
ENC_SW_RAW
D2
SP0504BAHT
R22
10k
R23
10k
R24
10k
R25
10k
R26
10k
R27
10k
C13
10n C0G
C14
10n C0G
C15
10n C0G

BOOST

U6
TPS61022
BOOST_EN
L1
1u
C19
10u X7R 16V
C20
22u X7R 10V
C21
22u X7R 10V
R31
732k
R32
100k
R33
100k

I2C

U3
TCA9546A
I2C_SDA
I2C_SCL
MUX_RST_N
HA_SDA
HA_SCL
HB_SDA
HB_SCL
R1
4.7k
R2
4.7k
C5
100n X8R
R7
10k

HAPTIC A

U1
DRV2605L
LRA_A_P
LRA_A_N
HAPTIC_TRIG
HAPTIC_EN
HA_REG
C1
1u X8R 25V
C2
1u X8R 25V
R3
4.7k
R4
4.7k
R11
100k
J2
LRA A

HAPTIC B

U2
DRV2605L
HB_SDA
HB_SCL
LRA_B_P
LRA_B_N
HB_REG
C3
1u X8R 25V
C4
1u X8R 25V
R5
4.7k
R6
4.7k
J3
LRA B

LED

SOS_LED_PWM
Q1
DMG2302U
LED_D
LED_K
R37
33
R38
33
R39
100k
D4
SOS red 0.5W

MECHANICAL

H1 M2
H2 M2
H3 M2
H4 M2

PWR_FLAG
PWR_FLAG
PWR_FLAG

Generated by aipcb (Claude-driven EDA)

aipcb

Sheet: /

File: outdoor_hmi.kicad_sch

Title: Outdoor Hub — HMI & Alarm Board (Board 2)

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.1		Id: 1/1